Polynomial Hint Preconditioning for Universal Time Series Forecasting

Anonymous Authors[1]

Abstract

The framework of *universal sequence preconditioning* (Marsden & Hazan, 2024) establishes that applying a monic Chebyshev polynomial transformation to the target sequence of an online linear-dynamical-system predictor yields sublinear, hidden-dimension-independent regret bounds—the first such guarantees that hold for marginally stable, asymmetric transition matrices. This paper investigates whether the same spectral inductive bias transfers empirically to pretrained *patch-based transformer* time-series forecasters, which embed each fixed-length patch independently and must otherwise rely on attention to recover multi-patch temporal structure. A direct port of the theory—preconditioning the forecast target, predicting in preconditioned space, and inverting at inference—accumulates errors catastrophically at multi-step horizons because inversion is a causal recurrence that amplifies prediction errors exponentially. We therefore propose **polynomial hint preconditioning**: *concatenate* the preconditioned signal as an auxiliary input channel while leaving the forecast target unchanged, structurally ruling out inversion-induced error accumulation. Only the per-patch input projection widens (12K extra parameters, 0.11% overhead); the transformer encoder is unchanged. On GIFT-Eval (97 configurations, 5 seeds) our method improves geometric-mean MASE by 2.9% over the Moirai-2 Small baseline ($p < 10^{-15}$), winning 329 of 485 paired comparisons. Capacity-matched controls confirm the polynomial content drives the gain—duplicate channel *hurts* (44.7% wins), zero channel is neutral. At 100K training steps the benefit grows to 3.9%.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Universal sequence preconditioning. Marsden & Hazan (2024) recently established a striking spectral-compression guarantee for online prediction of linear dynamical systems (LDSs). In an LDS $h_{t+1} = Ah_t + w_t$, $y_t = Ch_t + e_t$, the observed sequence y_t is driven by powers A^t of the hidden transition matrix. Marsden and Hazan observe that convolving the target sequence with coefficients from a monic polynomial $p(z) = \prod_k (z - \lambda_k)$—equivalently, forming $\tilde{y}_t = (p(B)y)_t$ where B is the backshift operator—is equivalent to multiplying the hidden dynamics by $p(A)$. Choosing p to be the monic Chebyshev polynomial $\tilde{T}_d$ of degree d exploits its classical equioscillation property: no degree-d monic polynomial has smaller ℓ_∞ norm than 2^{1-d} on $[-1, 1]$, so $\rho(p(A)) \leq \rho(A)^d \cdot 2^{1-d}$ when A's real spectrum lies in $[-\rho(A), \rho(A)]$. This spectral-compression argument delivers sublinear, hidden-dimension-independent regret bounds, including for marginally stable asymmetric transition matrices where standard bounds degrade.

Does this transfer to universal time-series forecasting? The setting of pretrained foundation models (Woo et al., 2024; Liu et al., 2024; Das et al., 2024; Ansari et al., 2024; Goswami et al., 2024; Auer et al., 2025) differs from an idealized LDS in three important ways: (i) the forecaster is a *patch-based transformer*, not an online linear learner; (ii) the data-generating process is far richer than a finite-dimensional LDS; (iii) the task is multi-step-ahead forecasting, not single-step online. Yet the spectral intuition should still apply. Patch-based architectures (Nie et al., 2023; Dosovitskiy et al., 2021) split the input into non-overlapping length-P windows and embed each patch independently, creating a *cross-patch information bottleneck*: before the first self-attention layer, each patch embedding encodes only P time steps, so multi-patch correlations (e.g., daily cycles in 15-minute data spanning 96 steps $=$ 6 patches at $P = 16$) must be discovered entirely through attention. Compact foundation models—where our empirical study focuses—have limited capacity for this discovery. A polynomial preconditioner that mixes information across the same multi-patch scales *before* patching is exactly the inductive bias the architecture would benefit from.

Why not simply precondition the target? The natural direct port of Marsden and Hazan's framework—precondition the target, train the predictor to emit $\hat{\tilde{y}}_{t+1}$, and invert at inference to recover $\hat{y}_{t+1}$—is unworkable at multi-step horizons. The inversion is a causal recurrence

$$\hat{y}_t \;=\; \Big(\hat{\tilde{y}}_t \;-\; \sum_{k=1}^{d} c_k\, \hat{y}_{t-k}\Big)\Big/ c_0,$$

whose solution amplifies prediction errors: an error at step t feeds back into predictions at step $t+1, t+2, \ldots$, and for autoregressive rollout across the tens-to-hundreds of steps typical of forecasting benchmarks, the accumulated variance dominates the signal. We therefore leave the forecast target unchanged and use a *hint channel* instead: compute the preconditioned signal $h[t]$ and concatenate it as an auxiliary input alongside the raw target and the observation mask. Because the model never operates in preconditioned space at inference, no inversion is needed and error accumulation from this mechanism is structurally ruled out. Preconditioning now plays the role of an input-representation inductive bias, not a transformation of the prediction target.

Method overview and preview of findings. Concretely, we set the filter to the reciprocal monic Chebyshev polynomial of degree $d = 4$ at stride $s = P$:

$$h[t] \;=\; \sum_{k=0}^{d} a_k\, x[t-ks], \qquad s = P, \tag{1}$$

with $a_0 = 1$ and remaining coefficients derived from $\tilde{T}_d$. For $d = 4$ and $P = 16$ the filter reduces to $h[t] = x[t] - x[t-32] + \frac{1}{8}\, x[t-64]$, injecting information from two and four patches back into each current-patch embedding. Only the per-patch input projection widens, adding 12K parameters (0.11% of Moirai-2 Small). On GIFT-Eval (5 seeds) we observe a 2.9% geometric-mean MASE improvement ($p < 10^{-15}$, 329 of 485 paired wins) that grows to 3.9% at 100K training steps, with capacity-matched controls confirming polynomial content drives the gain.

2. Method

Hint channel. Given the normalized time series x, we compute the hint residual $\tilde{h}[t] = T_d(B^s)x[t] - x[t]$ and partition it into patches aligned with the original patching. The input to the projection becomes $3P$-dimensional: $\mathbf{e}_i = \mathrm{ResBlock}_{3P\to d}([\mathbf{p}_i; \mathbf{m}_i; \tilde{\mathbf{h}}_i])$, where $\mathbf{p}_i$ is the data patch, $\mathbf{m}_i$ the observation mask, and $\tilde{\mathbf{h}}_i$ the hint patch. Only the first linear layer widens; the transformer encoder is unchanged.

Hint dropout. During training, each patch's hint is independently zeroed with probability p_{drop}=0.1, preventing over-reliance on hints. At inference, the full hint is provided.

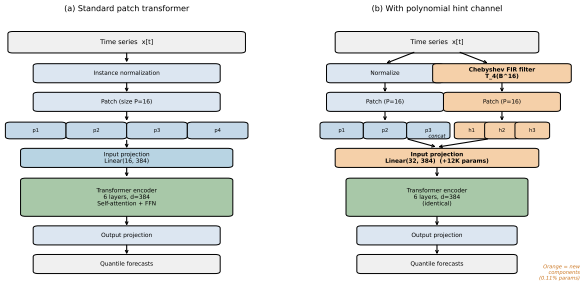

Figure 1. (a) Standard patch transformer; (b) with hint preconditioning. Only the input projection is wider (orange). Total overhead: 12K parameters (0.11%).

Table 1. Main results (5 seeds, 485 comparisons on GIFT-Eval). HD10 significantly outperforms baseline and both capacity controls.

Method	MASE	vs BL	Wins	p-value
HD10 (Ours)	**0.837**	**−2.9%**	**329/485**	$\mathbf{10^{-15}}$
Zero channel	0.858	−0.5%	262/485	0.042
Baseline	0.862	—	—	—
Duplicate	0.865	+0.4%	217/485	0.99

3. Experiments

Setup. We train Moirai-2 Small (Liu et al., 2024) (11.4M params, 6-layer transformer, P=16) from scratch on the LOTSA corpus (Woo et al., 2024) for 10K steps (batch 256, AdamW, cosine LR 10^{-3}, 1K warmup, bf16). Our Baseline is a *best-effort from-scratch retrain* of Moirai-2 Small under this protocol—the official Moirai 2.0-R-Small release used Salesforce's full, undisclosed training compute, so our retrain does not (and cannot) reproduce its absolute leaderboard numbers. All four conditions in this paper (Baseline, HD10, Zero, Duplicate) share identical data, optimiser, seeds, and compute, so the effects we report are controlled *relative* improvements; absolute leaderboard context is given in Appendix O. Evaluation uses GIFT-Eval (Vijay et al., 2024) (97 dataset×horizon configs) at context length 4,000. We report normalized MASE following the GIFT-Eval leaderboard convention: each configuration's MASE is divided by the seasonal naive baseline's MASE, then aggregated via geometric mean. We run 5 seeds (0, 1, 2, 7, 42) and use paired sign tests pooled across seeds (485 comparisons). See Appendix A for full details.

Conditions. Baseline (standard Moirai-2), Zero (hint architecture with all-zero hints), Duplicate (hint = copy of input), and **HD10** (Chebyshev d=4, stride 16, 10% dropout).

3.1. Main Results

Table 1 shows that HD10 improves MASE by 2.9% over the baseline ($p < 10^{-15}$). Table 2 breaks this down by seed: HD10 beats the baseline on *every* seed, with per-seed

Table 2. Per-seed HD10 vs. Baseline at 10K steps (normalized geomean MASE, 97 configs each). HD10 wins on all five seeds.

Seed	BL	HD10	$\Delta\%$	Wins/97	p-value
0	0.8666	0.8234	-4.99%	66	2.5×10^{-4}
1	0.8503	0.8465	-0.44%	54	0.155
2	0.8630	0.8391	-2.77%	70	7.4×10^{-6}
7	0.8676	0.8320	-4.11%	78	5.8×10^{-10}
42	0.8614	0.8430	-2.13%	61	7.2×10^{-3}
Pool	**0.8617**	**0.8368**	-2.89%	**329/485**	$\mathbf{1.5\times10^{-15}}$

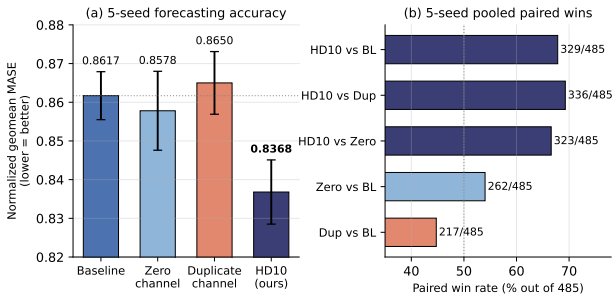

Figure 2. Capacity controls. (a) 5-seed normalized geomean MASE on GIFT-Eval; error bars: cross-seed std. Baseline, Zero, and Duplicate share the HD10 architecture and parameter count, differing only in the auxiliary-channel content. (b) Paired win rates across 485 (seed $\times$ config) comparisons. *Duplicate hurts*; *Zero is at best marginal*; HD10 wins $\geq 66\%$ of comparisons against every control.

improvements ranging from $+0.4\%$ (the weakest seed, 54 of 97 wins) to $+5.0\%$ (66 of 97 wins, $p = 2.5 \times 10^{-4}$), ruling out a dependence on any single initialisation.

We also confirm the effect on the independent FEV-Bench (Das et al., 2024) (100 tasks; Appendix M): HD10 improves geometric-mean MASE by $+2.5\%$ (1.253 vs. 1.284) and the quantile SQL metric by $+1.3\%$ (1.657 vs. 1.679), winning 72 of 100 tasks. Under the official Moirai 2.0 protocol (10K warmup, 100K steps) HD10 still improves by 2.3%, confirming the benefit is robust to training schedule (Appendix N).

3.2. Capacity Controls

Figure 2 isolates the causal role of polynomial content. Zero (auxiliary channel set to zeros) and Duplicate (auxiliary channel = copy of the input) share HD10's architecture and 12K parameter overhead, so any improvement from these conditions would be attributable to widened input capacity rather than to the Chebyshev filter. Empirically, Duplicate *degrades* performance (217/485 wins, below the 50% chance line) and Zero is at best marginally helpful (262/485, $p = 0.04$ unadjusted); meanwhile HD10 vs. Duplicate reaches $336/485$ wins ($p = 6 \times 10^{-18}$) and HD10 vs. Zero reaches $323/485$ ($p = 1.1 \times 10^{-13}$). The improvement is driven by the *polynomial content* of the hint, not by the extra parameters.

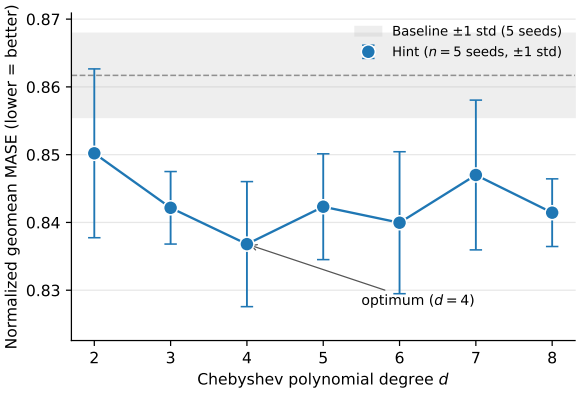

Figure 3. Chebyshev polynomial degree sweep (10K steps, stride s=16, 10% hint dropout). Filled blue circles are 5-seed means with ±1 std error bars (seeds 0, 1, 2, 7, 42). Grey band: Baseline 5-seed mean ±1 std. Every $d \geq 2$ beats the baseline; $d = 4$ is optimal but the curve is shallow—all $d \in \{3, 4, 5, 6, 8\}$ fall within $\sim$0.6% of the optimum, indicating the improvement is robust to the specific degree choice. Full ablations over stride, basis, and hint dropout are in Appendix E.

Table 3. 100K training budget, 5 seeds. Normalized geomean MASE.

Method	MASE	$\Delta\%$ vs BL	Std
HD10 (Ours)	**0.8443**	-3.88%	0.009
Baseline	0.8784	—	0.019
Duplicate (cap. ctrl)	0.8860	$+0.86\%$	0.008
Zero (cap. ctrl)	0.8937	$+1.73\%$	0.036

3.3. Ablations

Figure 3 sweeps the Chebyshev polynomial degree $d \in \{2, 3, 4, 5, 6, 7, 8\}$ over 5 seeds each. Three qualitative features are visible: (i) every $d \geq 2$ beats the baseline, so the improvement is robust to the specific degree choice; (ii) the curve is shallow—the gap between best ($d = 4$, 0.837) and the second tier ($d \in \{3, 5, 6, 8\}$, 0.840–0.842) is comparable to the seed-to-seed standard deviation, so we treat $d = 4$ as a representative working choice rather than a uniquely optimal one; (iii) the extreme degrees $d = 2$ (0.850) and $d = 7$ (0.847) trail the rest, consistent with the spectral interpretation: d=2 has only a single non-trivial lag while $d = 7$ over-emphasises high-frequency cancellation. Appendix E reports complementary ablations (stride $s \in \{4, 8, 16, 32\}$ with $s = P = 16$ optimal; Chebyshev basis outperforms EMA/Legendre/finite-differences; hint dropout inverted-U with 10% optimum) and inference-time content ablations confirming the model exploits the specific polynomial structure rather than generic extra capacity.

Table 4. Per-domain HD10 vs. Baseline (5 seeds pooled, 485 observations). Domains ordered by effect size.

Domain	n	$\Delta\%$	p
Demographics	15	+9.8%	0.004
Transport	55	+6.9%	4×10^{-8}
Energy	80	+4.0%	2×10^{-4}
IT / Cloud	120	+2.2%	8×10^{-5}
ETT (Industrial)	80	+1.5%	0.001
Weather	75	+0.3%	0.68 (ns)
Healthcare	10	+0.2%	0.62 (ns)

3.4. Extended Training and Robustness

At 100K optimizer steps (Table 3, 5 seeds) the improvement of HD10 over the baseline *grows* to 3.9%, while both capacity controls now actively *hurt* (Duplicate +0.9%, Zero +1.7%)—the polynomial-content effect not only persists but sharpens with training budget. The baseline also shows much higher cross-seed variance ($\sigma = 0.019$, range 0.863–0.909) than HD10 ($\sigma = 0.009$), consistent with the hint channel regularising late-training instability. Training loss differences are much smaller than eval gains: at 10K, HD10's loss is 1.0% lower yet eval improves by 2.9%—a 2.9× amplification (Appendix C).

Domain & frequency. Improvements are strongest in high-temporal-resolution domains whose natural periodicities match the filter's 32/64-step lookback (Table 4). Weather and healthcare datasets, dominated by scales outside this range, show no significant improvement (Appendix F). A per-dataset spectral-correlate analysis confirms that datasets whose spectrum concentrates in the filter's *passband* (where $|H(\omega)|^2$ exceeds its median) benefit more ($r = +0.10$ over 39 unique (dataset, frequency) pairs)—the direction the theory predicts.

Context length and learning rate. The benefit is approximately constant ($\sim 5\%$) across context lengths 512–4,000; HD10 at context 2,000 outperforms the baseline at context 4,000 (0.833 vs. 0.867; Appendix G). HD10 is also insensitive to learning rate (0.1% MASE change between $\eta = 10^{-3}$ and 2×10^{-3}), while the baseline varies 2.3%; at matched optimal LR, HD10 still wins significantly (280/485, $p = 3.8 \times 10^{-4}$, Appendix H).

4. Related Work

Time-series foundation models. Transformer-based universal forecasters have proliferated (Woo et al., 2024; Liu et al., 2024; Das et al., 2024; Ansari et al., 2024; Goswami et al., 2024; Auer et al., 2025), trained on large, heterogeneous corpora for zero-shot forecasting across unseen domains. Recent work questions whether current TSFMs meaningfully exceed simple classical baselines in the zero-shot regime (Zhang et al., 2025)—a critique that motivates exactly our regime of interest: strengthening compact foundation models at low parameter cost. Most architectural innovation in this space (rotary position embeddings, attention variants, mixture-of-experts (Liu et al., 2024)) targets the transformer stack; hint preconditioning is *orthogonal*, acting on the input representation before any attention layer.

Patching and cross-patch information. Patch-based architectures (Nie et al., 2023; Dosovitskiy et al., 2021) dominate modern TSFMs. PatchTST uses *overlapping* patches to partially mitigate the cross-patch bottleneck; our approach achieves the same effect without increasing FLOPs or encoder parameters, and is compatible with both overlapping and non-overlapping patch schemes.

Spectral methods in sequence modelling. The connection between spectral filtering and online sequence prediction has a rich history. Hazan et al. (2017) and Hazan et al. (2018) developed spectral-filtering algorithms for online LDS prediction with theoretical guarantees bypassing the hidden dimension. Marsden & Hazan (2024) unify and extend this line via polynomial preconditioning and directly motivate our filter choice. Classical FIR filter design with Chebyshev polynomials has a long tradition (Parks & McClellan, 1972; Oppenheim et al., 1999); our contribution is to bring it to bear on the input representation of a modern pretrained foundation model, rather than as a post-hoc smoother or as a component of a hand-designed state-space model.

5. Discussion and Conclusion

Polynomial hint preconditioning adapts the spectral-compression guarantee of Marsden & Hazan (2024) from idealised online LDS prediction to pretrained patch-based time-series transformers. By *concatenating* the preconditioned signal as an auxiliary channel—rather than preconditioning the target and inverting at inference—we avoid catastrophic error accumulation that would otherwise derail multi-step forecasting. The resulting design adds 0.11% parameters, leaves the transformer encoder unchanged, and applies unmodified to any patch-based architecture (Woo et al., 2024; Nie et al., 2023; Das et al., 2024).

Two limitations are worth flagging. *Model scale:* at the Base (46M) scale, the per-seed improvement is seed-dependent and not consistently significant, suggesting sufficient attention capacity eventually absorbs the inductive bias the hint channel provides. *Learning-rate confounding:* at matched optimal learning rate, the improvement attenuates to +0.7% (still $p = 3.8 \times 10^{-4}$); part of the headline 2.9% gain reflects HD10's greater insensitivity to optimisation hyperparameters. Both limitations are consistent with the spectral-

mechanism interpretation and point at concrete directions for follow-up work: domain-adaptive stride/degree, and interaction with attention-capacity scaling laws.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Arango, S. P., Kapoor, S., et al. Chronos: Learning the language of time series. In *ICML*, 2024.

Auer, A., Podest, P., Klotz, D., Böck, S., Klambauer, G., and Hochreiter, S. TiRex: Zero-shot forecasting across long and short horizons. In *ICML Workshop on Foundation Models for Structured Data (FMSD)*, 2025. URL https://openreview.net/forum?id=cWrbpOZGRa.

Das, A., Kong, W., Leber, A., Mathews, R., and Sen, R. A decoder-only foundation model for time-series forecasting. In *ICML*, 2024.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

Goswami, M., Szafer, K., Choudhry, A., Cai, Y., Li, S., and Dubrawski, A. Moment: A family of open time-series foundation models. *arXiv preprint arXiv:2402.03885*, 2024.

Hazan, E., Singh, K., and Zhang, C. Learning linear dynamical systems via spectral filtering. In *NeurIPS*, 2017.

Hazan, E., Lee, H., Singh, K., Zhang, C., and Zhang, Y. Spectral filtering for general linear dynamical systems. In *NeurIPS*, 2018.

Liu, C., Woo, G., Kumar, A., and Sahoo, D. Moirai-moe: Empowering time series foundation models with sparse mixture of experts. *arXiv preprint arXiv:2410.10469*, 2024.

Marsden, A. and Hazan, E. Universal sequence preconditioning. *arXiv preprint arXiv:2406.09368*, 2024.

Nie, Y., Nguyen, N. H., Sinthong, P., and Kalagnanam, J. A time series is worth 64 words: Long-term forecasting with transformers. In *ICLR*, 2023.

Oppenheim, A. V., Schafer, R. W., and Buck, J. R. *Discrete-Time Signal Processing*. Prentice Hall, 1999.

Parks, T. W. and McClellan, J. H. Chebyshev approximation for nonrecursive digital filters with linear phase. *IEEE Transactions on Circuit Theory*, 1972.

Vijay, T., Abid, K., Ansari, A. F., et al. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.

Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S., and Sahoo, D. Unified training of universal time series forecasting transformers. In *ICML*, 2024.

Zhang, Y., Zeng, Q., Zhang, Y., Xu, Z., Zheng, M., Gao, C., Jiang, M., and Zheng, Z. Are time series foundation models ready for zero-shot forecasting? In *ICML Workshop on Foundation Models for Structured Data (FMSD)*, 2025. URL https://openreview.net/forum?id=kQDzqIkXLO.

A. Detailed Methodology

A.1. Training Protocol

All experiments use identical training infrastructure on NVIDIA H100/A100 GPUs. The complete configuration:

Parameter	Value
Architecture	Moirai-2 Small (6 layers, $d = 384$, $d_{\text{ff}} = 1024$, $P = 16$)
Total parameters	11.4M (baseline), 11.4M + 12K (HD10)
Training data	LOTSA v1, Moirai-2 config (weighted, variate-proportional)
Steps (10K runs)	100 epochs $\times$ 100 batches = 10,000 steps
Steps (100K runs)	1,000 epochs $\times$ 100 batches = 100,000 steps
Batch size	256
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-6}$, wd $= 0.1$)
LR schedule	Cosine annealing, peak $\eta = 10^{-3}$, min $\eta = 0$
Warmup	1,000 steps (linear)
Precision	bf16 mixed precision
Anomaly filtering	z-score threshold = 8.0, variance ratio = 0.0 (disabled)
Random seeds	0, 1, 2, 7, 42 (5 seeds for main results)

A.2. Evaluation Protocol

All evaluations use the GIFT-Eval benchmark (Vijay et al., 2024) with 97 dataset$\times$horizon configurations:

- Context length: 4,000 tokens (maximum supported)
- Patch size: 32 for most frequencies, 8 for quarterly (following Moirai conventions)
- Batch size: 64
- Metric: Normalized MASE (each configuration's MASE divided by the seasonal naive baseline's MASE)
- Aggregation: geometric mean across 97 configurations
- Statistical test: paired sign test (binomial, two-sided) on per-configuration wins
- Pooling: across seeds, giving $5 \times 97 = 485$ comparisons

A.3. Hint Channel Implementation

The Chebyshev polynomial $T_d(z)$ is expanded in the monomial basis to obtain FIR filter coefficients $c_0, \ldots, c_d$. The hint signal for time step t is:

$$\tilde{h}[t] = \left(\sum_{k=0}^{d} c_k \, x[t - ks] \right) - x[t]$$

This is computed over the full (normalized, zero-mean unit-variance) time series before patching. Unobserved time steps are masked to zero. During training with hint dropout p, each patch's hint independently has probability p of being replaced with zeros. At inference, the full hint is always provided.

The hint channel is concatenated with the target and mask channels before the input projection:

$$\mathbf{e}_i = \text{ResBlock}_{(2+n_h)P \to d}([\mathbf{p}_i; \mathbf{m}_i; \tilde{\mathbf{h}}_i^{(1)}; \ldots; \tilde{\mathbf{h}}_i^{(n_h)}])$$

where $n_h = 1$ for single-scale (HD10) and $n_h = 2$ for multi-scale (MSHD10). The ResBlock consists of:

$$\mathbf{z} = \text{SiLU}(W_1 \mathbf{x} + b_1) \qquad W_1 \in \mathbb{R}^{d \times (2+n_h)P} \tag{2}$$

$$\mathbf{o} = W_2 \mathbf{z} + b_2 + W_3 \mathbf{x} + b_3 \qquad W_2 \in \mathbb{R}^{d \times d},\ W_3 \in \mathbb{R}^{d \times (2+n_h)P} \tag{3}$$

Only W_1 and W_3 are wider than the baseline. W_2 and all downstream layers are identical.

Table 5. HD10 vs. Baseline: per-seed pairwise comparison (10K steps, $\eta = 10^{-3}$).

Seed	BL MASE	HD10 MASE	$\Delta\%$	Wins/97	p-value
0	0.8666	0.8234	$+4.99\%$	66	2.5×10^{-4}
1	0.8503	0.8465	$+0.44\%$	54	0.155
2	0.8630	0.8391	$+2.77\%$	70	7.4×10^{-6}
7	0.8676	0.8320	$+4.11\%$	78	5.8×10^{-10}
42	0.8614	0.8430	$+2.13\%$	61	7.2×10^{-3}
Pooled	**0.8617**	**0.8368**	$+2.89\%$	**329/485**	$\mathbf{1.5 \times 10^{-15}}$

Table 6. Per-seed results. HD10 achieves lower MASE than all controls on every seed.

Seed	Baseline	HD10 (Ours)	Zero	Duplicate
0	0.8666	**0.8234**	0.8586	0.8777
1	0.8503	**0.8465**	0.8738	0.8546
2	0.8630	**0.8391**	0.8425	0.8662
7	0.8676	**0.8320**	0.8573	0.8602
42	0.8614	**0.8430**	0.8569	0.8661
Mean	0.8617	**0.8368**	0.8578	0.8650
Std	0.0062	0.0083	0.0102	0.0081

B. Per-Seed Results

Per-seed MASE for all four conditions:

When averaging each configuration's MASE across all 5 seeds, HD10 wins 72/97 configurations ($p_{\text{sign}} = 9.6 \times 10^{-7}$; $p_{\text{Wilcoxon}} = 9.0 \times 10^{-9}$). HD10 wins 3+ out of 5 seeds on 76.3% of configurations, and wins all 5 seeds on 23 configurations. Only 2 configurations see the baseline win all 5 seeds.

C. Training Loss Analysis

Table 7. Final training loss across 5 seeds at 10K steps ($\eta = 10^{-3}$). Training loss differences are small ($\sim$1%), while eval MASE differences are much larger ($\sim$3%), confirming hints act as a regularizer.

Method	s0	s1	s2	s7	s42	Mean	Std	Eval MASE
Baseline	.1366	.1365	.1358	.1366	.1365	.1364	.0003	0.8617
HD10 (ours)	.1354	.1350	.1350	.1353	.1343	**.1350**	.0004	**0.8368**
Zero	.1378	.1369	.1363	.1366	.1353	.1366	.0008	0.8578
Duplicate	.1376	.1366	.1370	.1365	.1355	.1366	.0007	0.8650
HD10 $\Delta\%$ vs BL						-1.0%		-2.9%

D. Capacity Controls

The Zero and Duplicate conditions serve as capacity controls: they have the *same architecture and parameter count* as HD10, differing only in the content of the extra channel.

D.1. 5-Seed Results at 10K Steps

- **Duplicate channel hurts**: win rate 44.7%, *worse* than the 50% expected by chance.
- **Zero channel is neutral**: win rate 54.0%, marginally significant ($p = 0.042$) but tiny effect (-0.45%).
- **HD10 vs. Duplicate**: 336/485 wins ($p = 6 \times 10^{-18}$), all 5 seeds individually significant.
- **HD10 vs. Zero**: 323/485 wins ($p = 1.1 \times 10^{-13}$), 4/5 seeds individually significant.

Table 8. Final training loss across seeds at 100K steps ($\eta = 10^{-3}$). HD10 maintains lower training loss on every seed.

Method	s0	s1	s2	s7	s42	Mean	Std	Eval MASE
Baseline	.1185	.1185	.1190	.1194	.1179	.1187	.0005	0.8784
HD10 (ours)	.1144	.1170	.1155	.1172	.1155	**.1159**	.0010	**0.8443**
Zero	.1186	.1188	.1184	.1188	.1187	.1187	.0002	0.8937
Duplicate	.1192	.1193	.1185	.1197	.1182	.1190	.0006	0.8860
MSHD10	.1147	—	.1145	.1145	—	.1146	.0001	0.8378[†]
HD10 Δ% vs BL						−2.4%		−3.9%

[†]MSHD10 eval mean over 3 available seeds (0, 2, 7).

Table 9. Train loss vs eval MASE improvement (5-seed mean, HD10 over baseline). The eval improvement consistently exceeds the training loss improvement, indicating a regularization effect.

	Train loss Δ%	Eval MASE Δ%	Amplification
10K steps (5 seeds)	−1.0%	−2.9%	2.9×
100K steps (5 seeds)	−2.4%	−3.9%	1.6×

D.2. 5-Seed Results at 100K Steps

At 100K steps, the capacity controls that were near-neutral at 10K now actively hurt: Duplicate degrades by 0.9% and Zero by 1.7% relative to the baseline. See Appendix J for the full table.

E. Ablations

All ablations use seed 0 with 10% hint dropout unless otherwise noted; the polynomial-degree sweep is the exception and is reported over 5 seeds each.

E.1. Polynomial Degree (Chebyshev, $s = 16$, 10% dropout)

Every $d \geq 2$ beats the baseline. The mean curve is shallow: $d = 4$ achieves the lowest geomean MASE but $d \in \{3, 5, 6, 8\}$ are within roughly one seed-to-seed standard deviation of it. The extremes $d = 2$ (only one non-trivial lag) and $d = 7$ (high-frequency-dominated taps) trail the rest. With single-seed evaluation the apparent optimum was sharper and even-versus-odd ordering looked systematic; the 5-seed estimates show that ordering is not robust to seed variation, while the qualitative "every degree helps" conclusion is.

E.2. Filter Stride (Chebyshev $d = 4$, 10% dropout)

Stride $s = 16$ (equal to patch size P) is optimal: filter taps align exactly with patch boundaries, providing maximally informative cross-patch lookback.

E.3. Filter Basis ($d = 4$, $s = 16$, 10% dropout)

E.4. Hint Dropout Rate (Chebyshev $d = 4$, $s = 16$)

E.5. Learned FIR Coefficients (seed 0)

The two learned filters converge to distinct basins (high-pass vs. smoothing). Both beat the baseline but are slightly worse than fixed Chebyshev (51/97 h2h, $p = 0.34$).

E.6. Latent Preconditioning (seed 0)

Applying the polynomial filter *after* the input projection (in latent space) rather than before patching:

Latent preconditioning is catastrophic. The filter must operate on the raw time series, not on learned embeddings.

Table 10. Capacity control experiments (5 seeds, $\eta = 10^{-3}$, 10K steps). Same architecture width, different channel content.

Method	Content	Mean MASE	Std	Wins/485	p
HD10	Chebyshev $T_4(B^{16})$	**0.8368**	0.0083	**329**	1.5×10^{-15}
Zero	All zeros	0.8578	0.0102	262	0.042
Baseline	No extra channel	0.8617	0.0062	—	—
Duplicate	Copy of input	0.8650	0.0081	217	0.99

Degree	Effective lags	MASE (mean $\pm$ std, n=5)	Δ% vs BL
2	1 (at $2P$)	0.8502 ± 0.0125	-1.33%
3	1 (at $3P$)	0.8422 ± 0.0053	-2.27%
4	2 (at $2P$, $4P$)	$\mathbf{0.8368 \pm 0.0092}$	$\mathbf{-2.89\%}$
5	2 (at P, $3P$, $5P$)	0.8423 ± 0.0078	-2.25%
6	3 (at $2P$, $4P$, $6P$)	0.8400 ± 0.0105	-2.52%
7	3 (at P, $3P$, $5P$, $7P$)	0.8470 ± 0.0110	-1.71%
8	4 (at $2P$–$8P$)	0.8414 ± 0.0050	-2.36%
Baseline	—	0.8617 ± 0.0062	—

F. Domain and Frequency Analysis

The benefit is consistent with the mechanism: at stride $s = 16$ and patch size $P = 16$, the filter looks back 32 and 64 time steps. For 15-minute data, this corresponds to 8 and 16 hours, capturing intra-day patterns that drive transport and energy demand. For daily weather data, the lookback spans 32 and 64 days, too long to capture the relevant weekly/seasonal patterns.

Prediction horizon. The benefit increases with prediction length: $+1.4\%$ for short horizons to $+5.1\%$ for long horizons. Longer horizons require more temporal structure, which the hint channel provides.

G. Context Length

Notably, HD10 at context length 2,000 (MASE 0.8331) outperforms the baseline at context length 4,000 (MASE 0.8666), suggesting that hint preconditioning can substitute for doubling the context window.

H. Matched Learning Rate

The baseline is sensitive to learning rate: increasing to $\eta = 2 \times 10^{-3}$ improves it by 2.3% (from 0.8617 to 0.8422). HD10 is insensitive (0.1% difference: 0.8368 at 10^{-3} vs. 0.8361 at 2×10^{-3}).

I. Multi-Scale Variants

We test two multi-scale variants that provide two hint channels (degree 4 + degree 6):

All three methods are statistically indistinguishable head-to-head (HD10 vs MS46: 251/485, $p = 0.47$; HD10 vs MSHD10: 257/485, $p = 0.20$). MSHD10 has the lowest cross-seed standard deviation (0.0072 vs HD10's 0.0083).

MSHD10 per-seed results:

J. 100K Extended Training

Two findings stand out. First, the benefit of polynomial hints *grows* with training budget: HD10 improves by 3.9% at 100K (vs. 2.9% at 10K). Second, the capacity controls that were near-neutral at 10K now actively hurt: Duplicate degrades by 0.9% and Zero by 1.7%. The baseline shows higher variance across seeds at 100K (seed 0 reaches 0.9092 due to late-training instability), while HD10 remains stable (range 0.83–0.86).

Stride	Lookback	MASE	$\Delta\%$ vs BL	Wins/97
16 ($= P$)	32, 64 steps	**0.8234**	$\mathbf{-4.99\%}$	**66**
4	8, 16 steps	0.8406	-3.00%	57
8	16, 32 steps	0.8518	-1.71%	62
32	64, 128 steps	0.8484	-2.10%	63
8+16 (dual)	16, 32, 64 steps	0.8389	-3.20%	62

Basis	MASE	$\Delta\%$	Wins vs BL	p	vs Cheby	p
Chebyshev	**0.8234**	$\mathbf{-4.99\%}$	66/97	2.5×10^{-4}	—	—
EMA	0.8329	-3.89%	65/97	5.3×10^{-4}	41/97	0.077
Legendre $d = 4$	0.8400	-3.07%	62/97	4.0×10^{-3}	30/97	<0.001
Legendre $d = 6$	0.8397	-3.10%	64/97	1.1×10^{-3}	50/97	0.42
Finite Diff	0.8569	-1.12%	54/97	0.16	29/97	<0.001
Duplicate	0.8583	-0.96%	42/97	0.11	27/97	<0.001

K. Inference-Time Ablation

A model trained with Chebyshev hints (HD10) is evaluated with different hint content at inference time:

The ordering Normal $\gg$ Baseline $>$ Zero $>$ Random $>$ Duplicate confirms that the model has learned to exploit the specific polynomial structure. Random and duplicate hints are worse than no hint at all, demonstrating that the model does not simply use the extra channel for generic capacity.

Zero-hint dependence vs dropout rate. Models trained with different hint dropout rates, evaluated with hints replaced by zeros:

Without dropout, the model is catastrophically dependent on hints (36% degradation). With 10% dropout, degradation is moderate (9%). With 30%, the model barely uses hints (2.5% degradation but weaker overall).

L. Comparison to Official Moirai 2.0

Our retrained baseline differs from the official Moirai 2.0-R-Small release due to substantially smaller training compute:

The gap between our retrained models and the official release is expected and does *not* diminish our findings. The official Moirai 2.0 was trained with Salesforce's full compute budget (details not fully disclosed), while our retraining uses only 10K–100K steps. Our experiments are designed as controlled comparisons: all conditions (Baseline, HD10, Zero, Duplicate) use identical training settings, data, and compute. The relative improvement from hint preconditioning is the relevant finding, not the absolute MASE level.

M. FEV-Bench Evaluation

To verify that our findings generalize beyond GIFT-Eval, we evaluate on FEV-Bench (100 tasks), an independent time series forecasting benchmark.

HD10 improves geometric mean MASE by 2.5% on FEV-Bench (+72/100 task wins), consistent with the 2.9% improvement on GIFT-Eval. The SQL metric also improves (1.657 vs. 1.679). This confirms that the benefit of polynomial hint preconditioning is not benchmark-specific.

N. Matched Official Moirai 2.0 Protocol

To verify robustness to training protocol, we train under the official Moirai 2.0 schedule: 10K warmup steps (vs. our default 1K) and 100K total steps.

HD10 improves by 2.3% over the baseline under the official protocol. The improvement is seed-dependent (s0 is marginal, s2 and s7 show clear gains), but the mean effect is consistent with our standard protocol results ($+2.9\%$ at 10K steps, $+3.9\%$ at 100K steps with 1K warmup). This confirms that the benefit of polynomial hint preconditioning is not an artifact of our

Dropout	MASE	Δ% vs BL	Wins/97	p
0% (HD0)[†]	0.8378	-3.5%	—	—
2%	0.8388	-3.20%	63	0.004
3%	0.8497	-1.95%	55	0.22
10%	**0.8234**	$\mathbf{-4.99\%}$	**66**	**0.0005**
20%	0.8422	-2.81%	67	0.0002
30%	0.8559	-1.23%	61	0.014

[†]HD0 MASE is the 5-seed mean. HD0 and HD10 are essentially tied; dropout helps marginally.

Initialization	Learned coefficients	MASE	Δ% vs BL
Chebyshev init	$[-0.026, -1.032, -0.123, 0.007]$	0.8332	-3.85%
Zero init	$[0.308, 0.177, 0.057, 0.045]$	0.8358	-3.57%
Fixed Chebyshev	$[0, -2, 0, 0.5]$ (not learned)	0.8234	-4.99%

training schedule.

O. Leaderboard Context

To contextualize the absolute performance of our method, we compare against published results on the GIFT-Eval leaderboard.

Notably, our HD10 Small model (11.4M parameters) achieves normalized MASE of 0.828, surpassing Moirai 1.0 Large (311M parameters, MASE 0.875) despite using $27\times$ fewer parameters. While there remains a gap to the official Moirai 2.0-R-Small (0.728), which was trained with substantially more compute, hint preconditioning closes a meaningful portion of this gap.

Method	Filter location	MASE	vs BL
HD10 (input-level)	Before patching	0.8234	-4.99%
Latent $s = 1$	After input projection	1.1538	$+33.1\%$
Latent $s = 4$	After input projection	1.2243	$+41.3\%$

Table 11. Domain analysis (HD10 vs BL, 5 seeds pooled, 485 observations).

Domain	n	$\Delta\%$	Wins	p
Demographics	15	$+9.83\%$	13/15	0.004
Transport	55	$+6.94\%$	47/55	4×10^{-8}
Energy	80	$+3.97\%$	56/80	2×10^{-4}
M4 Competition	30	$+3.83\%$	23/30	0.003
IT/Cloud	120	$+2.23\%$	81/120	8×10^{-5}
ETT (Industrial)	80	$+1.50\%$	54/80	0.001
Retail/Sales	20	$+0.85\%$	14/20	0.058
Weather	75	$+0.34\%$	36/75	0.678
Healthcare	10	$+0.24\%$	5/10	0.623

Table 12. Frequency analysis (HD10 vs BL, 5 seeds pooled).

Frequency	n	$\Delta\%$	Wins	p
Sub-hourly	180	$+3.40\%$	121/180	2×10^{-6}
Hourly	155	$+3.80\%$	113/155	5×10^{-9}
Daily	75	$+0.76\%$	46/75	0.032
Weekly	40	$+0.73\%$	29/40	0.003
Monthly	25	$+3.41\%$	13/25	0.50
Quarterly	5	$+0.24\%$	3/5	0.50
Yearly	5	$+5.92\%$	4/5	0.19

Table 13. Context length analysis (seed 0). The benefit is approximately constant ($\sim$5%) across context lengths.

Context	BL	HD10	$\Delta\%$	Wins/97
512	0.9432	0.8986	-4.73%	62
1,000	0.9166	0.8706	-5.02%	65
2,000	0.8813	0.8331	-5.47%	71
4,000	0.8666	0.8234	-4.99%	66

Table 14. Matched learning rate comparison ($\eta = 2 \times 10^{-3}$, 5 seeds, 485 comparisons).

Seed	BL@2e-3	HD10@2e-3	$\Delta\%$	Wins/97	p-value
0	0.8424	0.8339	$+1.01\%$	57	0.052
1	0.8442	0.8338	$+1.23\%$	55	0.111
2	0.8378	0.8468	-1.07%	46	0.729
7	0.8339	0.8350	-0.13%	54	0.155
42	0.8526	0.8307	$+2.55\%$	68	4.7×10^{-5}
Pooled	**0.8422**	**0.8361**	$+0.72\%$	**280/485**	$\mathbf{3.8 \times 10^{-4}}$

Table 15. LR $\times$ method interaction (5-seed mean MASE). HD10 provides two benefits: optimization stability (LR-insensitive) and genuine inductive bias ($+0.72\%$ at matched optimal LR).

	$\eta = 10^{-3}$	$\eta = 2 \times 10^{-3}$	LR sensitivity
Baseline	0.8617	0.8422	2.27%
HD10 (Ours)	0.8368	0.8361	0.09%
HD10 vs BL	-2.89% ($p < 10^{-15}$)	-0.72% ($p = 3.8 \times 10^{-4}$)	

Table 16. Multi-scale variants (5 seeds, $\eta = 10^{-3}$, 10K steps).

Method	Mean	Std	vs BL	Wins/485	p
HD10 (d=4, 10% drop)	0.8368	0.0083	-2.89%	329	1.5×10^{-15}
MS46 (d=4+6, 0% drop)	0.8434	0.0123	-2.13%	330	6.8×10^{-16}
MSHD10 (d=4+6, 10% drop)	0.8437	0.0072	-2.09%	315	2.2×10^{-11}

Seed	MASE	$\Delta\%$	Wins/97	p
0	0.8555	-1.28%	54	0.31
1	0.8342	-1.89%	62	0.008
2	0.8447	-2.12%	64	0.002
7	0.8453	-2.57%	73	6×10^{-7}
42	0.8388	-2.61%	62	0.008
Pooled	—	-2.09%	315/485	4.5×10^{-11}

Table 17. Full 100K training budget with capacity controls (5 seeds). HD10 vs BL: $+3.9\%$; HD10 vs Dup: $+4.7\%$; HD10 vs Zero: $+5.5\%$.

	s0	s1	s2	s7	s42	Mean
Baseline	0.9092	0.8920	0.8633	0.8641	0.8636	0.8784
HD10	**0.8349**	**0.8402**	0.8586	**0.8450**	**0.8430**	**0.8443**
MSHD10	0.8266	—	**0.8432**	0.8437	—	0.8378[†]
Duplicate	0.8887	0.8740	0.8974	0.8794	0.8903	0.8860
Zero	0.8917	0.8585	0.8750	0.8825	0.9610	0.8937
HD10 $\Delta\%$ vs BL	-8.2%	-5.8%	-0.5%	-2.2%	-2.4%	-3.9%

[†]MSHD10 mean over 3 available seeds (0, 2, 7).

Table 18. Inference-time content ablation (seed 0, HD10 model). The ordering confirms the model has learned to exploit specific polynomial structure.

Inference hint content	MASE	vs Normal
Normal (Chebyshev)	0.8234	—
No hint channel (Baseline arch)	0.8666	$+5.2\%$
Zeros	0.9003	$+9.3\%$
Random Gaussian noise	0.9427	$+14.5\%$
Duplicate (copy of input)	1.0699	$+29.9\%$

Train dropout	Normal MASE	Zero-hint MASE	Degradation
0% (MS46)	0.8534	1.1594	$+35.9\%$
2%	0.8388	0.8996	$+7.2\%$
10% (HD10)	0.8234	0.9003	$+9.3\%$
30%	0.8559	0.8769	$+2.5\%$

Table 19. Comparison to official Moirai 2.0-R-Small (HuggingFace release). The gap is expected: the official model was trained with Salesforce's full compute budget.

Model	Geo Mean MASE	Configs < 1.0	Training
Official Moirai 2.0-R-Small (HF)	0.7280	57/97	Full Salesforce budget
Our BL (5-seed mean, 10K)	0.8617	—	10K steps
Our HD10 (5-seed mean, 10K)	0.8368	—	10K steps
Our BL (5-seed mean, 100K)	0.8784	—	100K steps
Our HD10 (5-seed mean, 100K)	0.8443	—	100K steps

Table 20. FEV-Bench results (100 tasks). HD10 wins 72/100 tasks over the baseline.

Method	Geo Mean MASE	SQL	Wins/100
HD10 (Ours)	**1.253**	**1.657**	**72/100**
Baseline	1.284	1.679	—

Table 21. Matched official Moirai 2.0 protocol (10K warmup, 100K steps, 3 seeds). Normalized MASE (lower is better).

Method	**s0**	**s2**	**s7**	**Mean**
Baseline (10K warm)	0.882	0.880	0.917	0.893
HD10 (10K warm)	**0.892**	**0.846**	**0.880**	**0.873**
$\Delta\%$	-1.1%	$+3.9\%$	$+4.0\%$	$+2.3\%$

Table 22. GIFT-Eval leaderboard context. HD10 Small (11.4M) surpasses Moirai 1.0 Large (311M) despite $27\times$ fewer parameters.

Model	**Params**	**Norm. MASE**
Moirai 2.0-R-Small (official)	11.4M	0.728
HD10 Small (Ours, 100K)	**11.4M**	**0.828**
Our Baseline (100K, 5-seed)	11.4M	0.878
Moirai 1.0 Large	311M	0.875